

HUSNAIN GHAFAR

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Profile

Electrical Engineer with a strong interest in power systems, renewable energy, and SCADA. Eager to apply academic knowledge to real-world projects, collaborate with experienced professionals, and grow in a dynamic team environment.

Education

COMSATS University Islamabad, Lahore Campus

Lahore, Pakistan

Bachelor of Science in Electrical Engineering

2021 - 2025

Professional Experience

TurkPak International Pvt. Ltd

Lahore, Pakistan

Junior Electrical Engineer

Sep 2025 – Present

- Conducted load flow and short-circuit studies for low- and medium-voltage distribution networks improving feeder voltage profile compliance by $\approx 10\%$ using ETAP and PSS[®]E.
- Performed feeder load forecasting and performance analysis using Synergi Electric for multiple distribution feeders.

NATIONAL ENGINEERING SERVICES PAKISTAN (NESPAK)

Lahore, Pakistan

Intern

Aug 2024 – Sep 2024

- **Renewable Energy:** Evaluation of grid-connected solar PV systems and SCADA.
- **Technical Expertise:** Acquired hands-on experience in designing and troubleshooting solar PV systems.

PAK ELEKTRON LIMITED (PEL)

Lahore, Pakistan

Summer Intern

Jul 2024 – Aug 2024

- **Design & Testing:** Worked on the complete design and testing of power transformers, ensuring compliance with industry standards. Assisted in performance evaluations and optimization to improve efficiency and reliability.

KARACHI AUTOMATION

Lahore, Pakistan

Internship Trainee

Jun 2024 – Jul 2024

- **Automation:** Programmed PLC logic and configured SCADA interfaces for industrial automation panels, improving system monitoring efficiency and reducing manual intervention during operations.

Academic Projects

IoT-Based Cardiac Health Monitoring System

- The final-year project was developed for real-time health monitoring of drivers. Spring-2025
- Designed and implemented a system integrating IoT and machine learning to monitor vital signs in real-time, detect anomalies, and trigger alerts for timely medical intervention.

Power Distribution and Transformer Optimization using ETAP

Spring-2025

- Designed and simulated a 40-house LV distribution system in ETAP. Evaluated load variations with EV integration and optimized transformer sizing to ensure voltage stability and system efficiency.

Industrial Power System Analysis

Fall - 2024

- Performed analysis of industrial power systems using ETAP. Focused on load flow analysis, short circuit studies, power stability, motor starting analysis, and capacitor placement for optimal system performance.

Pitch Angle Control System of a Satellite

Spring-2023

- Simulated and analyzed a satellite's pitch angle control system using MATLAB as part of a university semester project on satellite attitude control.

DC Motor Speed Control (PID Controller)

Fall - 2023

- Implemented a PID controller to regulate DC motor speed under varying load conditions, achieving stable performance using real-time feedback mechanisms.

Power Generation by Footsteps

Fall - 2022

- Implemented a system to harness energy from footsteps. This project demonstrated a potential solution for small-scale renewable energy generation in public spaces.

Extracurricular & Volunteer Experience

IDP Ambassador

- Represented IDP at campus events, guiding students on study abroad opportunities and organizing promotional activities.

Departmental Promotion Society

- Member/Event Organizer

IEEE Student Branch

- Head of Photography

IEEE Robotics and Automation Society (RAS)

- Team Member (Social Media & Photography)

Alkhidmat Foundation

- Volunteer

IEEE RAS RoboWar Competition

- Participant & Management Team Member

Honors and Awards

- Runner-Up – RoboWar Competition, IEEE RAS
- Team Member – IEEE RAS Media Team
- Participant – National Engineering Olympiad, GIKI
- Participant – Chess Tournament at COMSATS University

CERTIFICATIONS

- Lean Six Sigma White Belt Certification
- ETAP Software Certification
- Solar Energy System Design
- MATLAB and Machine Learning Onramp
- AutoCAD Certification (DigiSkills Pakistan)
- Communication and Soft Skills Training (DigiSkills Pakistan)

Technical Skills & Interests

- **Software & Tools:** PSS/E, MATLAB, ETAP, AutoCAD, PVSyst, Microsoft Office
- **Technical Skills:** Power System Analysis & Design, PLC Programming, SCADA, Solar System Design, Electrical Load Estimation, Control Systems, Fault Diagnosis & Troubleshooting
- **Interests:** Renewable Energy Integration, Control Systems, System Optimization
- **Hobbies:** Photography, Badminton, Chess